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Ridgeway et al.

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(54) **SHALLOW ELECTRICAL PROTECTION
DEVICE (GFCL, AFCL, AND AFCL/GFCL)
SYSTEM AND METHOD**

(58) **Field of Classification Search**

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H01R 24/78; H01R 25/006;

(Continued)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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Padro Kenny; Title: Compact latching mechanism for switched
electrical device; entire specification and drawings (Year: 2013).*

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Friedrich LLP

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G01R 11/04 (2006.01)

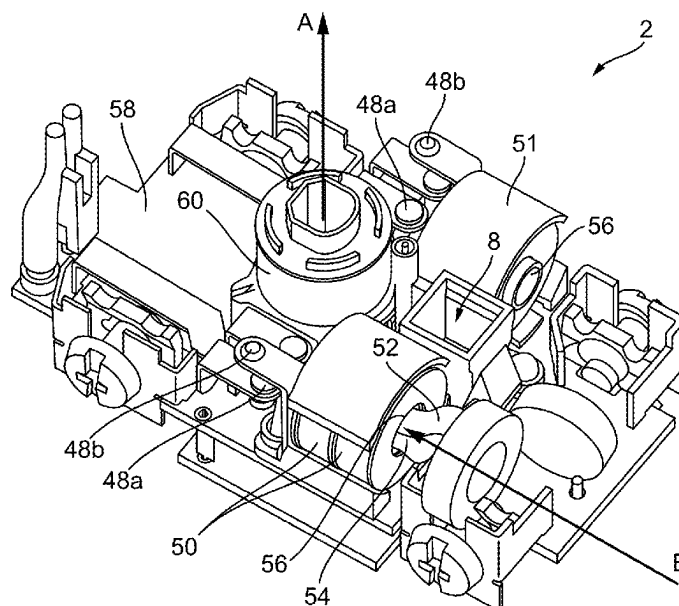
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CPC **H01R 13/7135** (2013.01); **G01R 11/04**
(2013.01); **H01H 71/505** (2013.01);
(Continued)

(57) **ABSTRACT**

A device may include a housing including a face plate. A
device may include a plurality of sensing cores each con-
figured to receive a current flow through a center cavity, the
current flow defining a current flow direction through the
center cavity, wherein the current flow direction is parallel
to the face plate.

14 Claims, 27 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/698,690, filed on Mar. 18, 2022, now Pat. No. 11,557,862, which is a division of application No. 16/212,141, filed on Dec. 6, 2018, now Pat. No. 11,309,667.

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(51) **Int. Cl.**

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H01R 24/76 (2011.01)
H01R 24/78 (2011.01)
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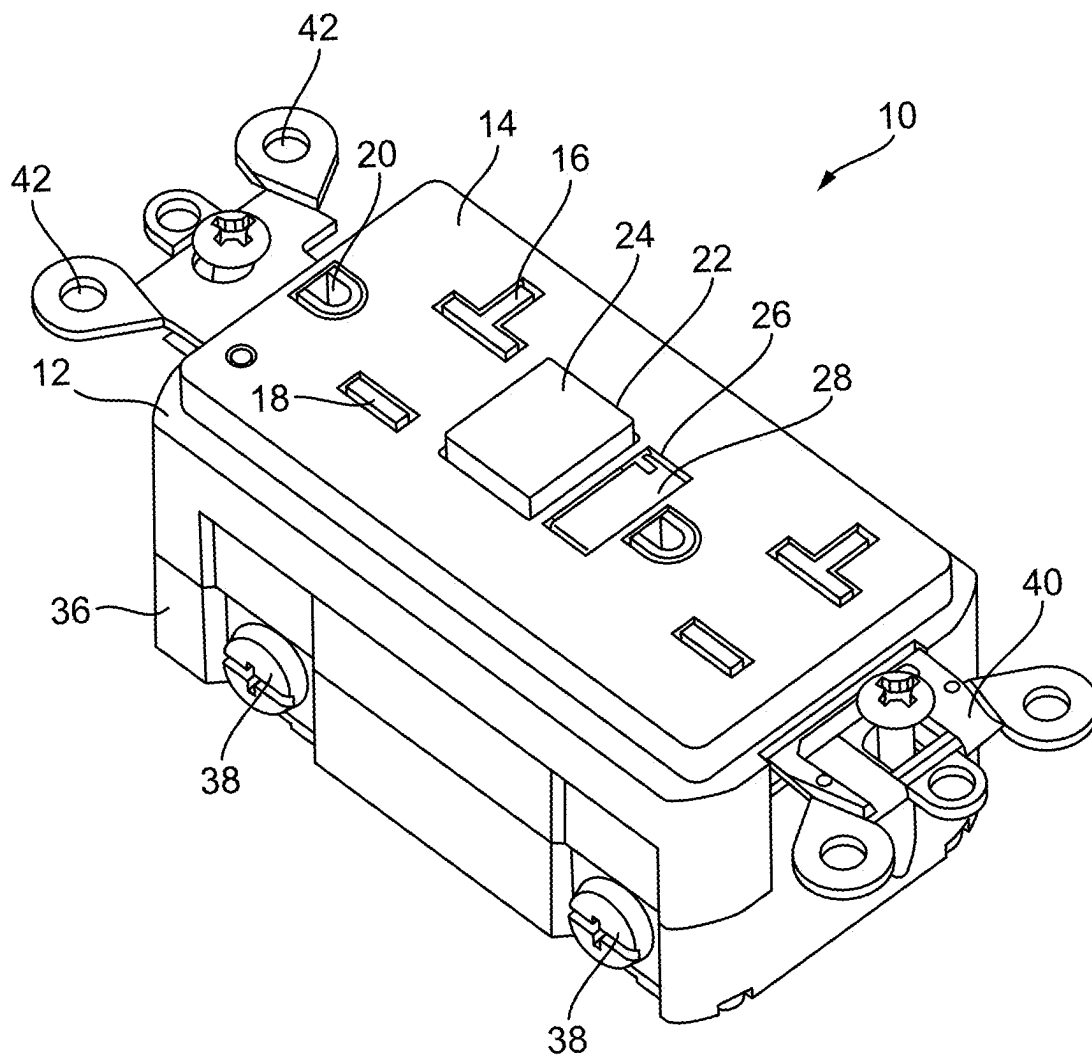


FIG. 1

FIG. 2

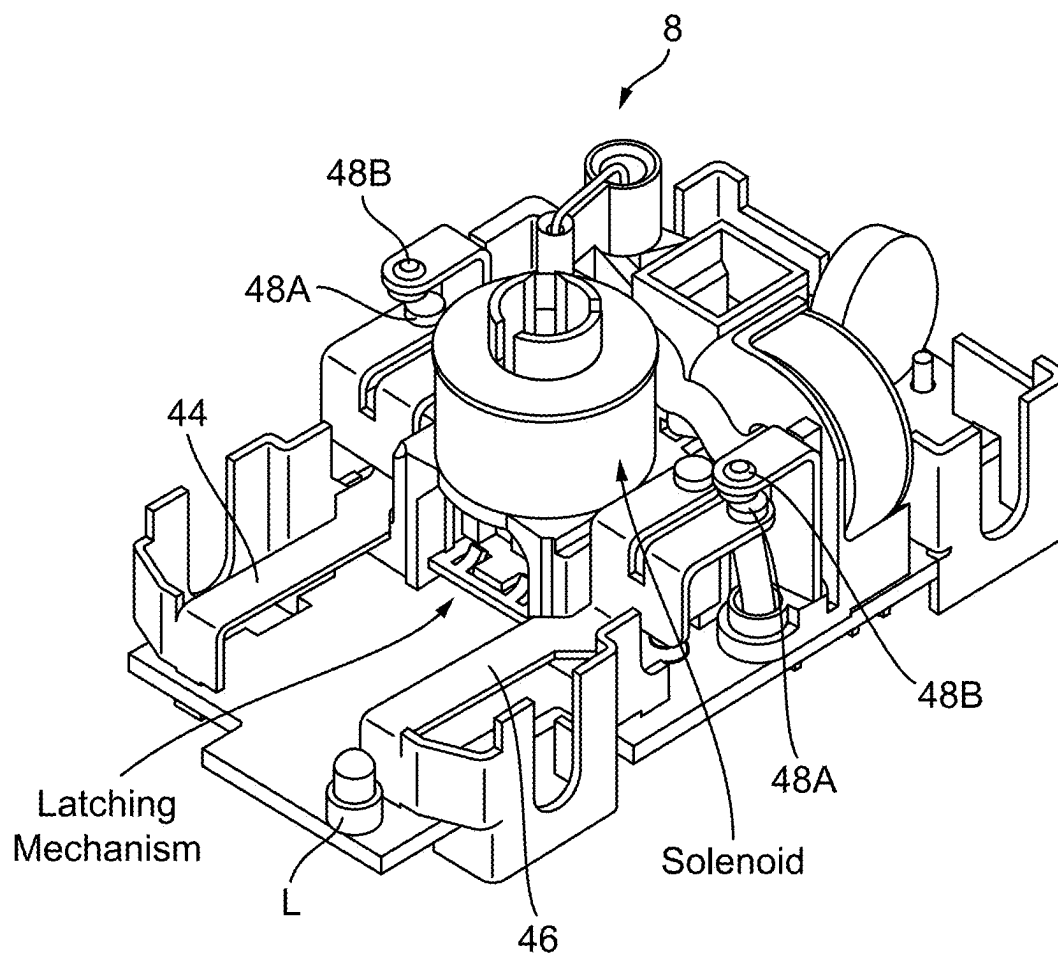


FIG. 3

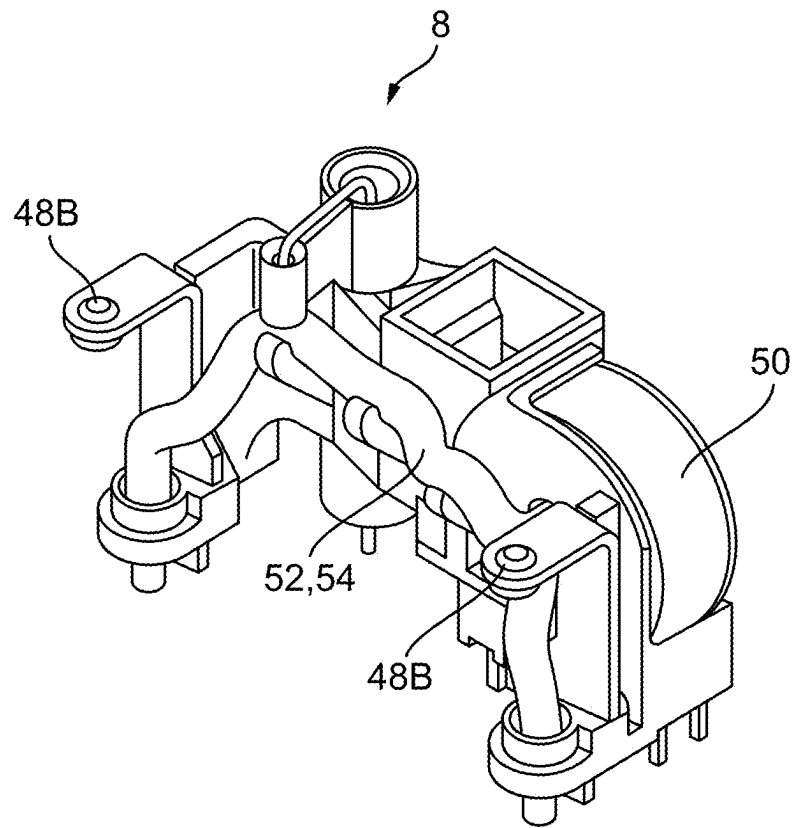


FIG. 4A

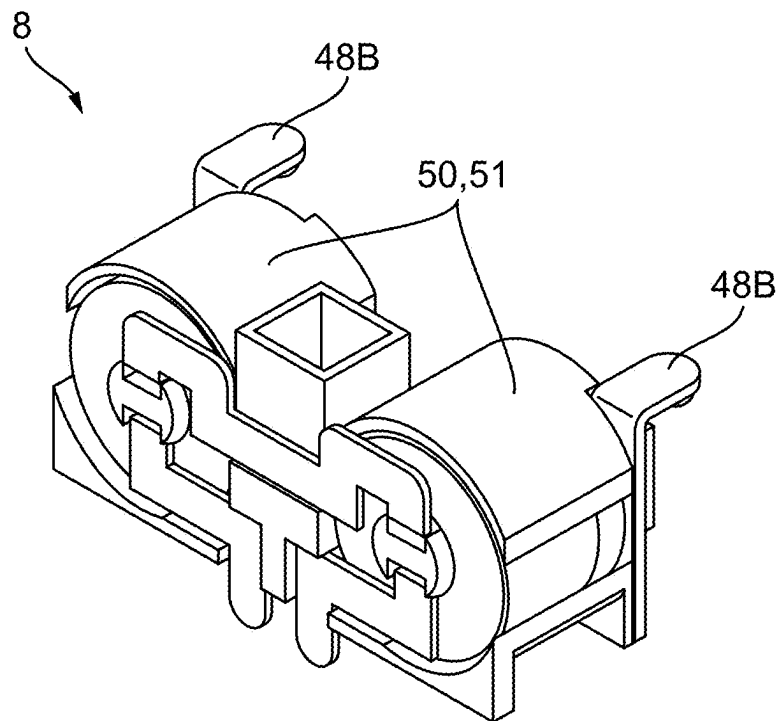


FIG. 4B

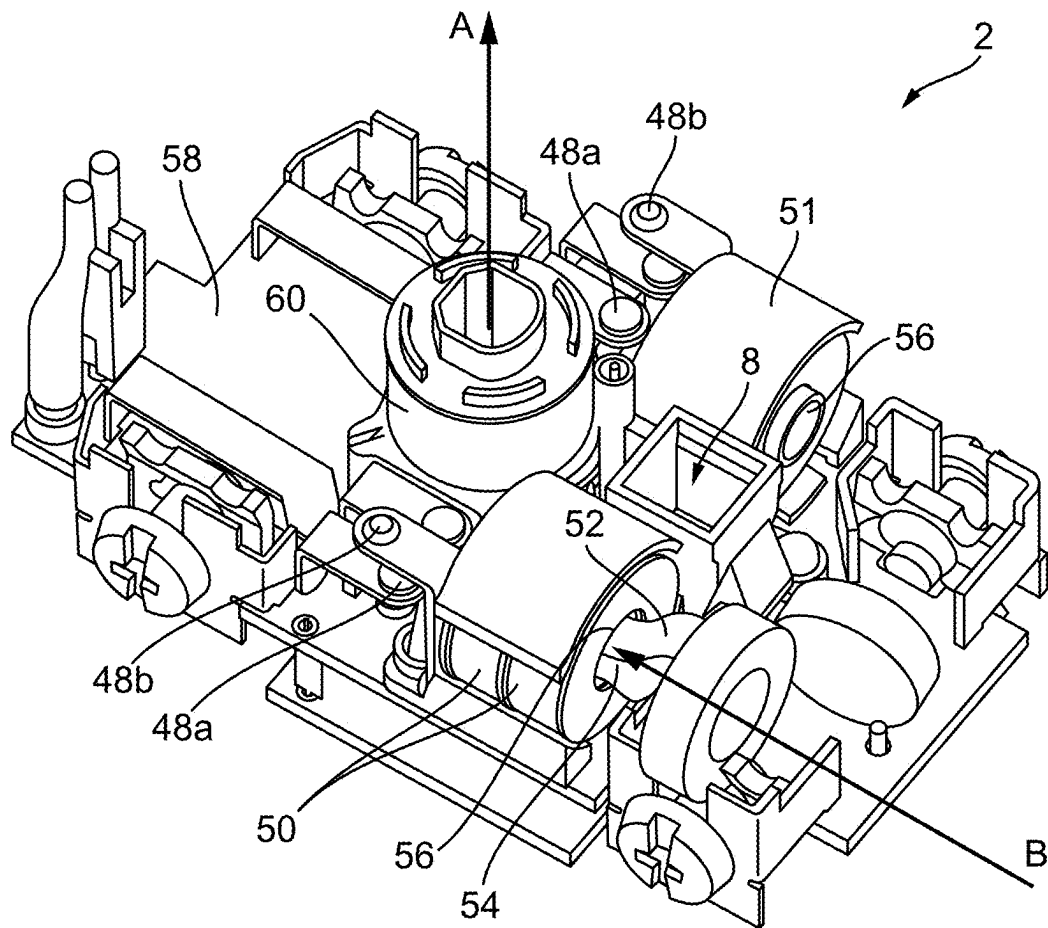


FIG. 5A

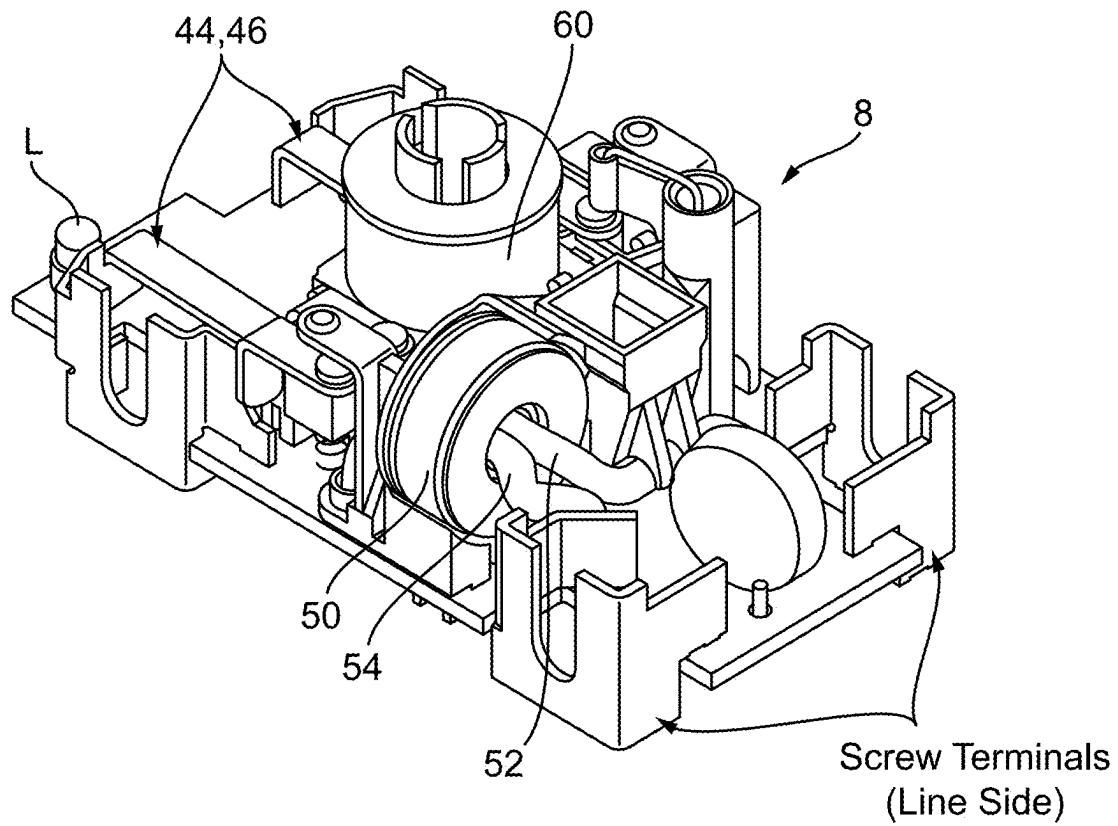


FIG. 5B

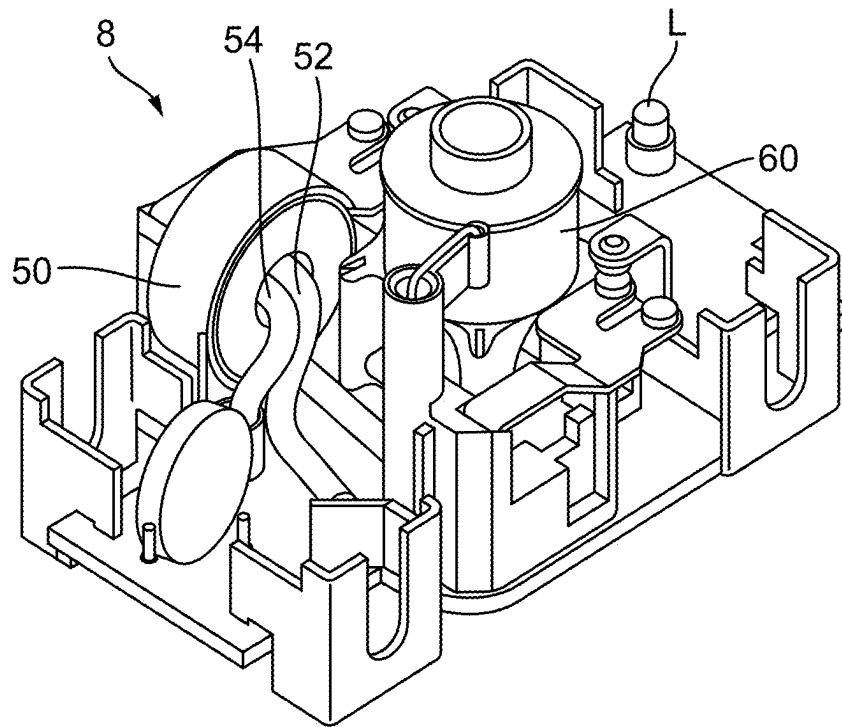


FIG. 5C

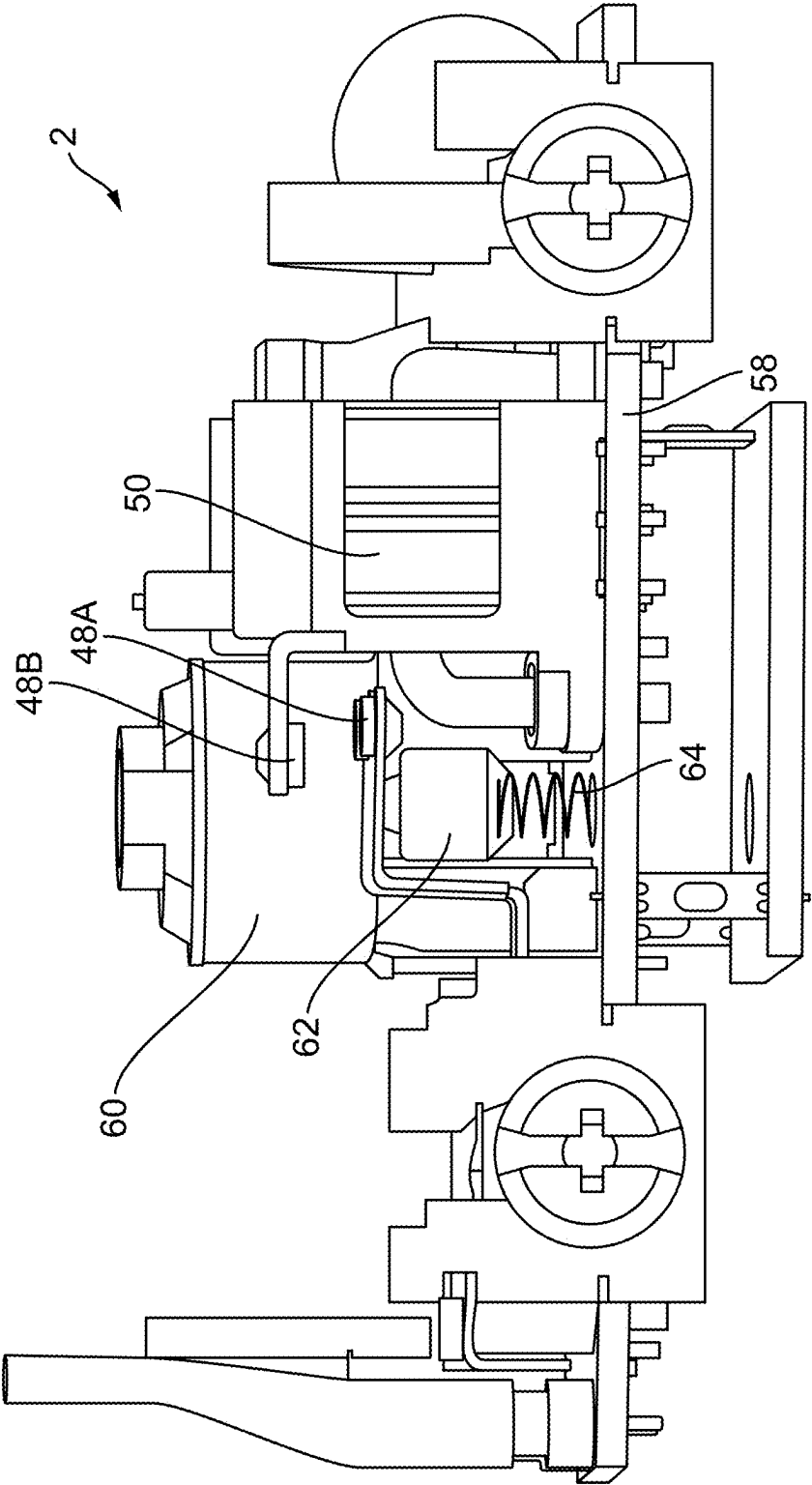


FIG. 6A

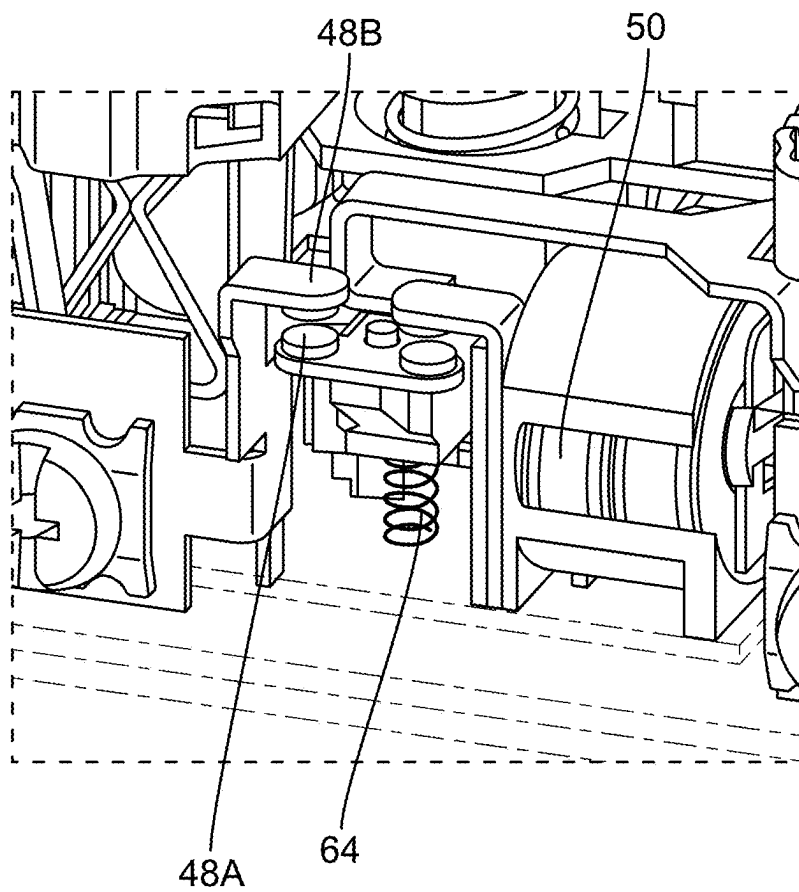


FIG. 6B

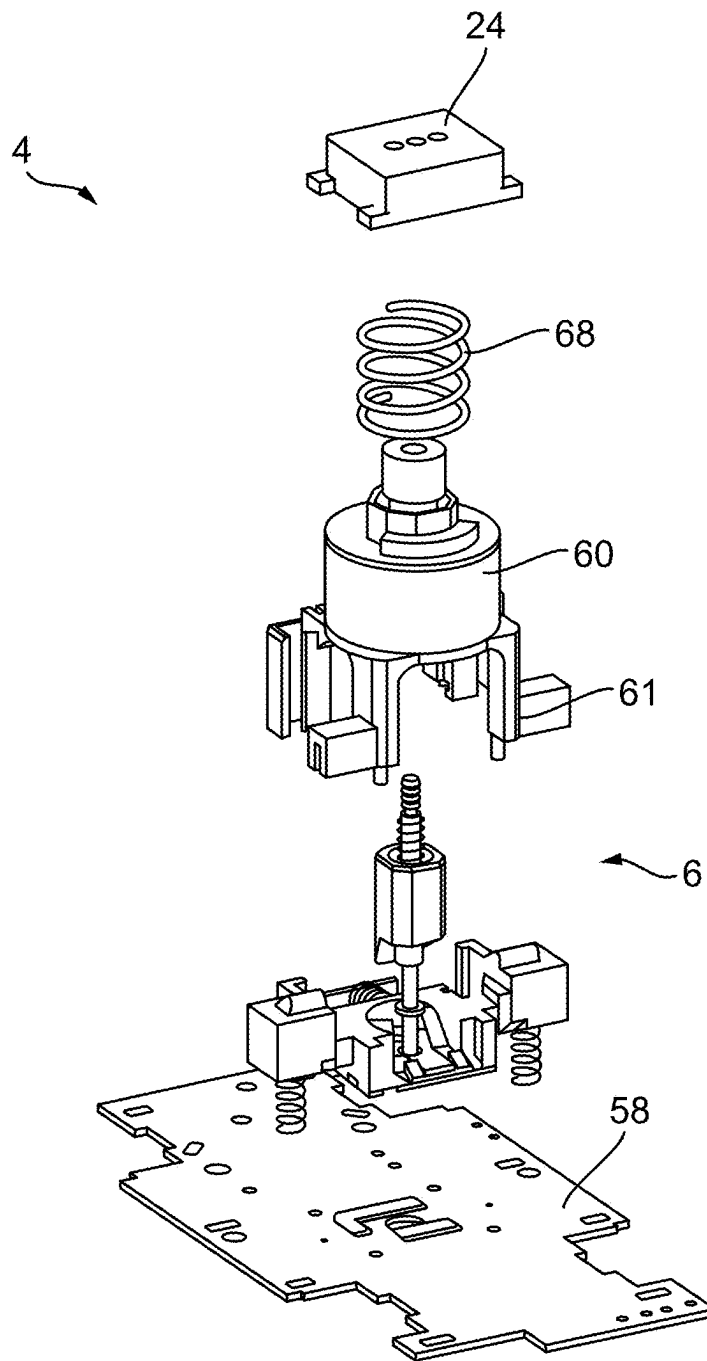


FIG. 7

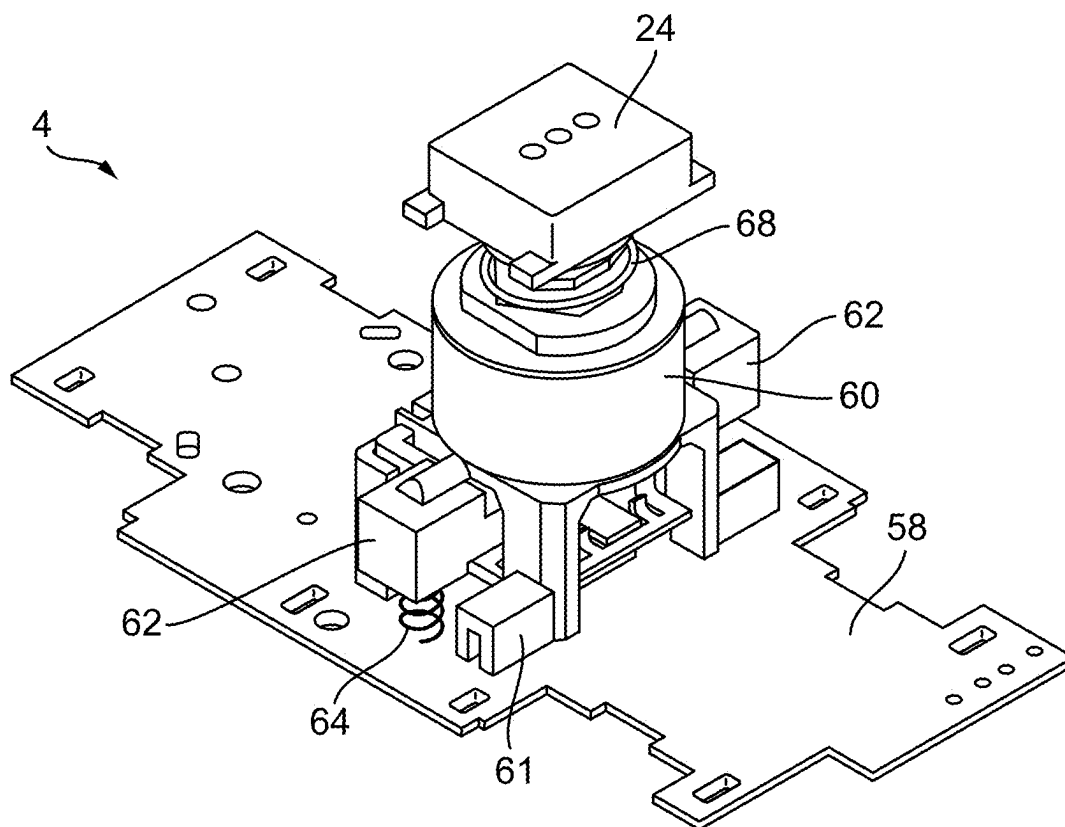


FIG. 8

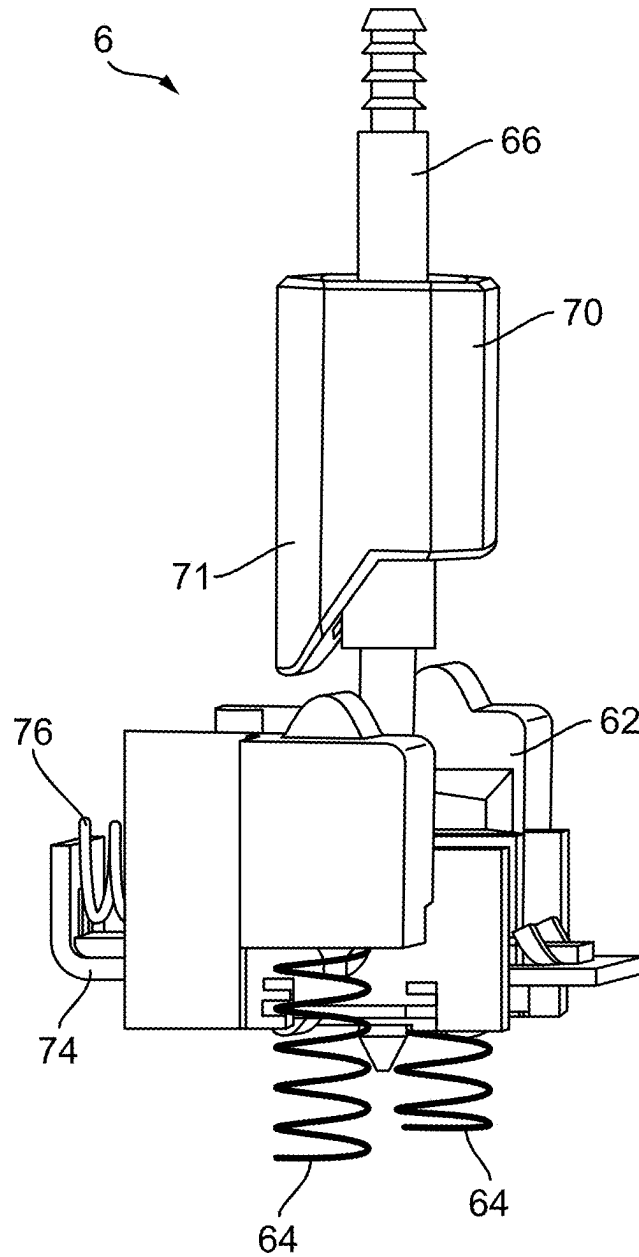


FIG. 9

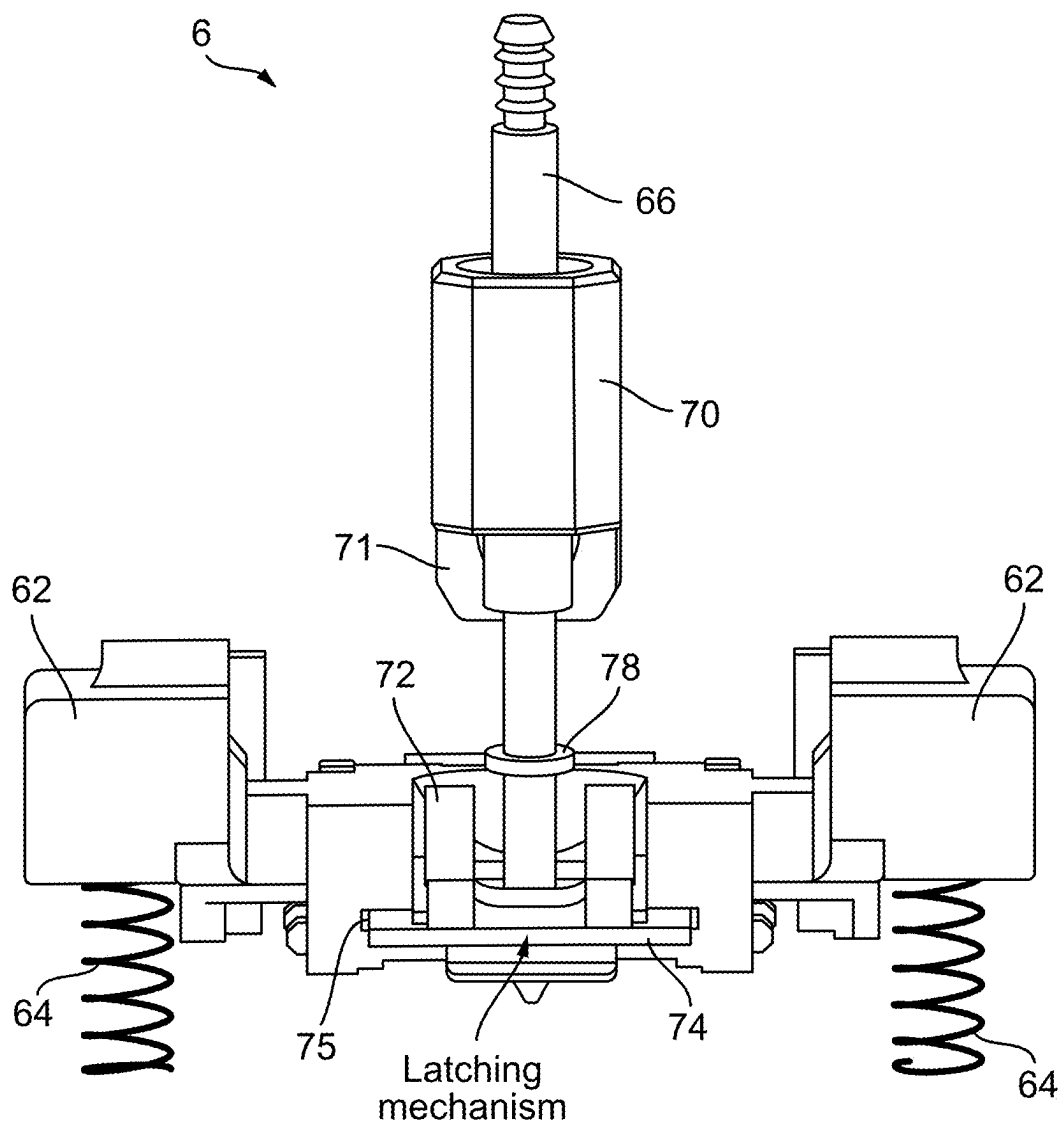


FIG. 10

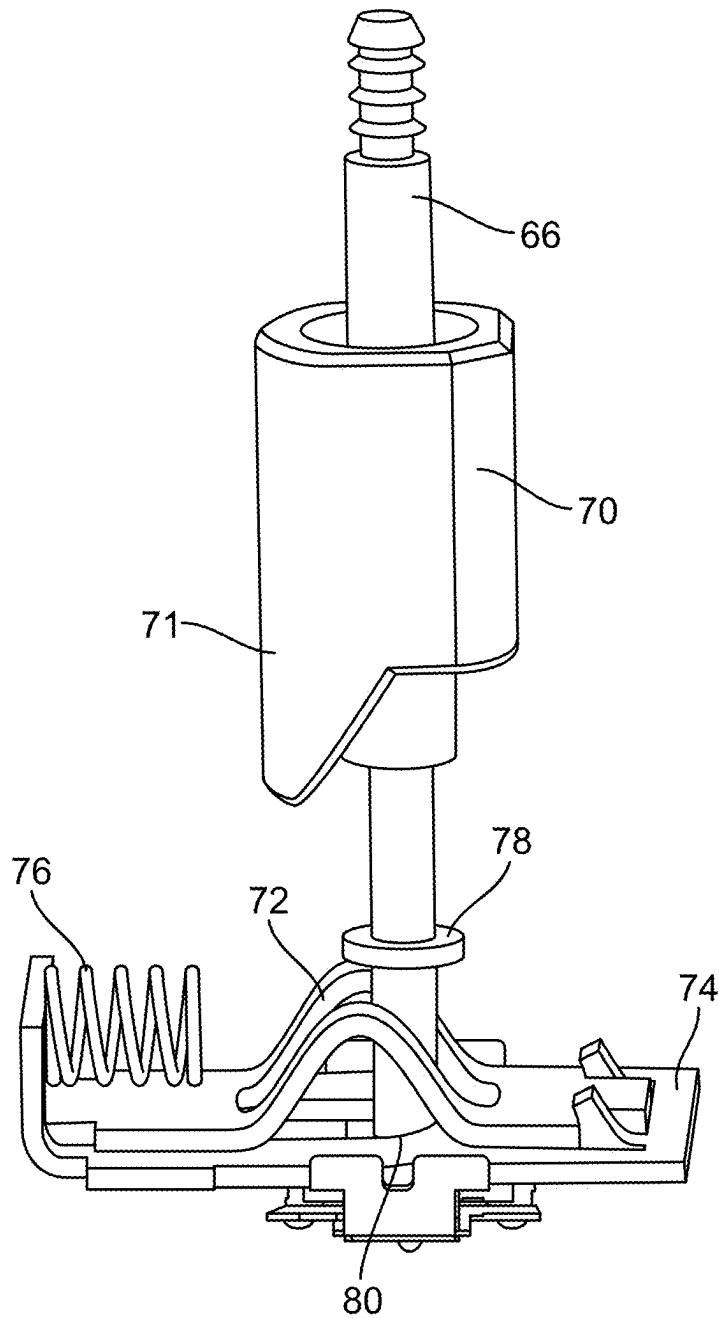


FIG. 11

Two-piece latching

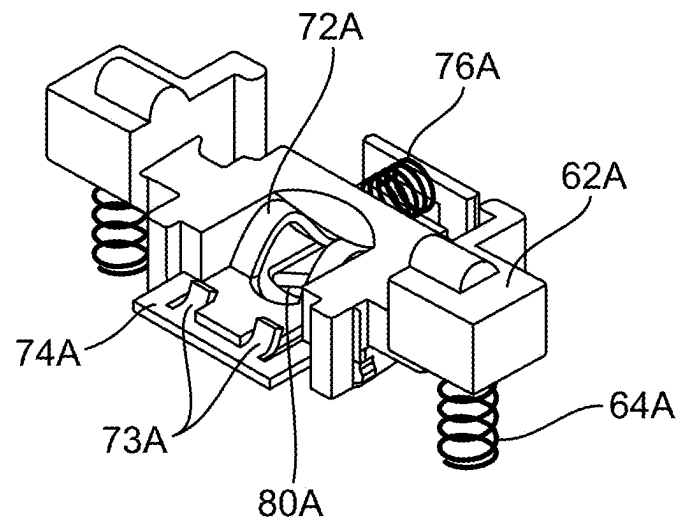


FIG. 12A

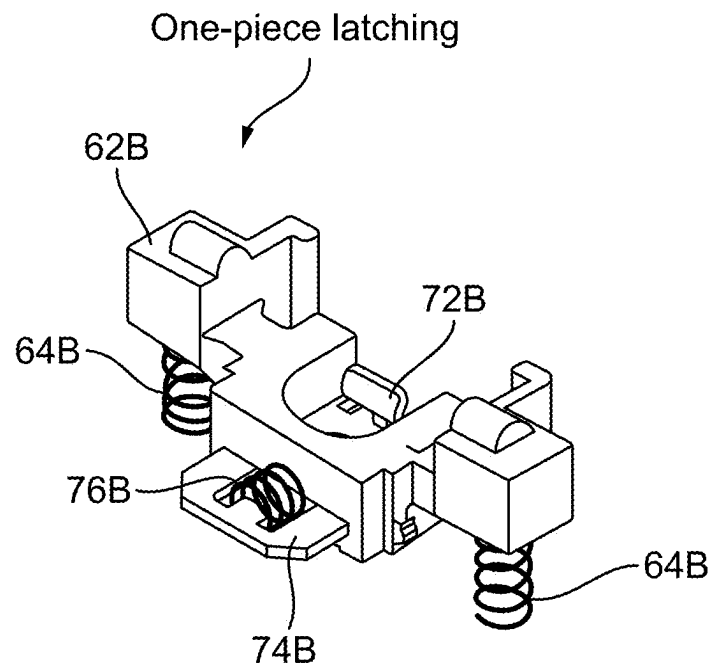


FIG. 12B

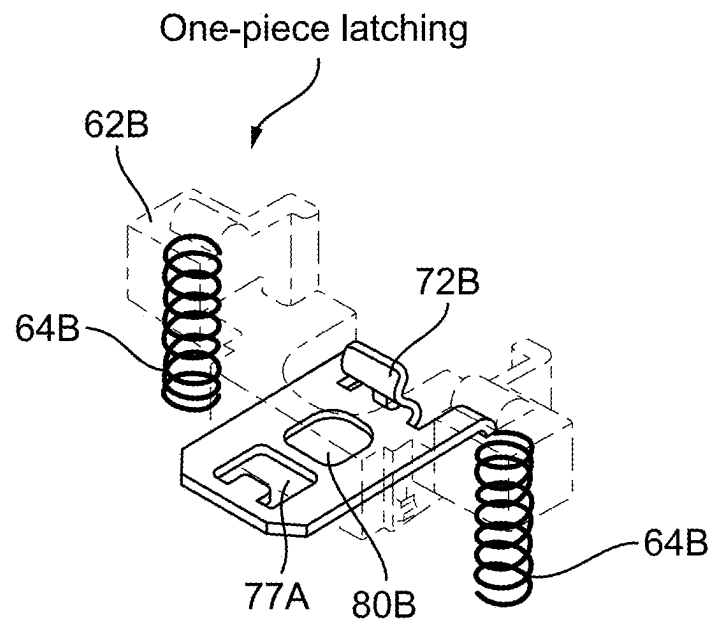


FIG. 12C

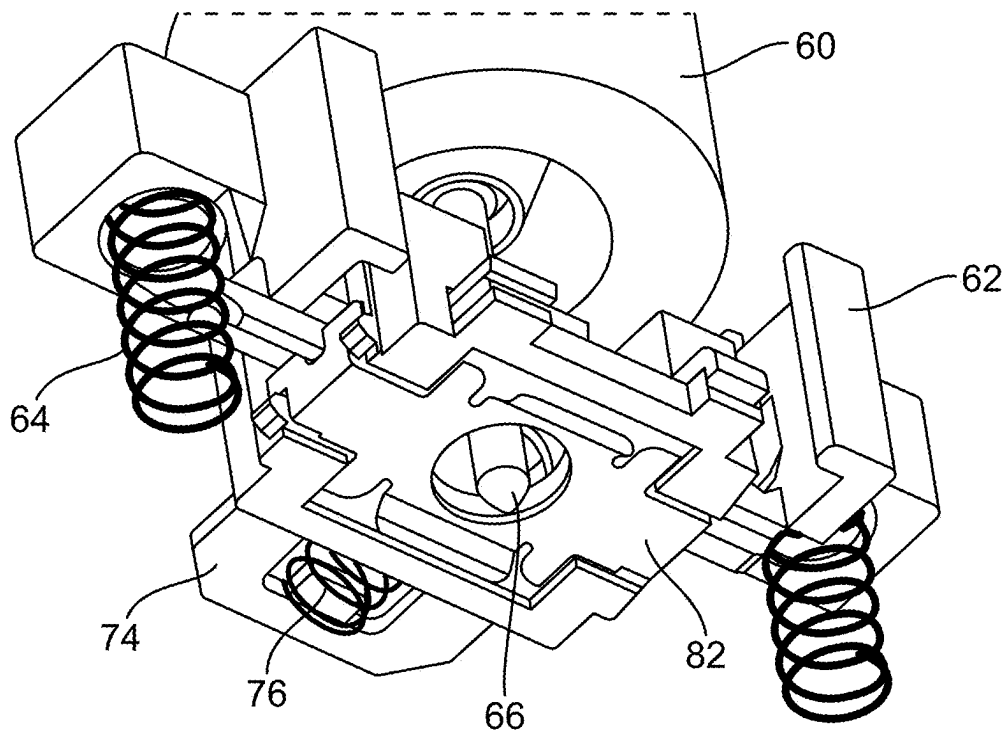


FIG. 13

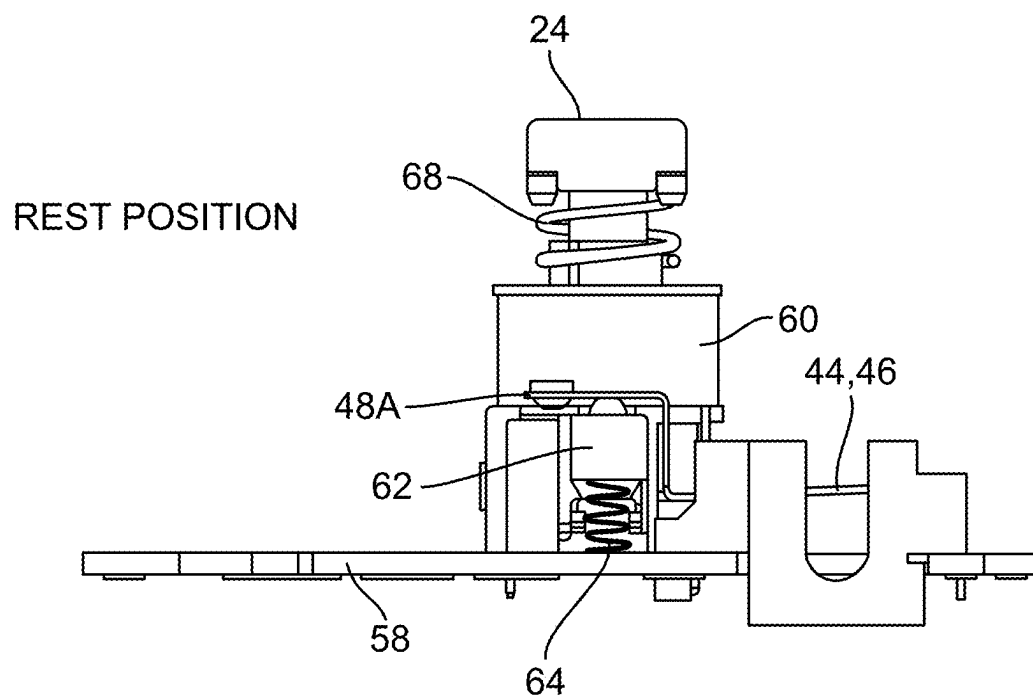


FIG. 14A

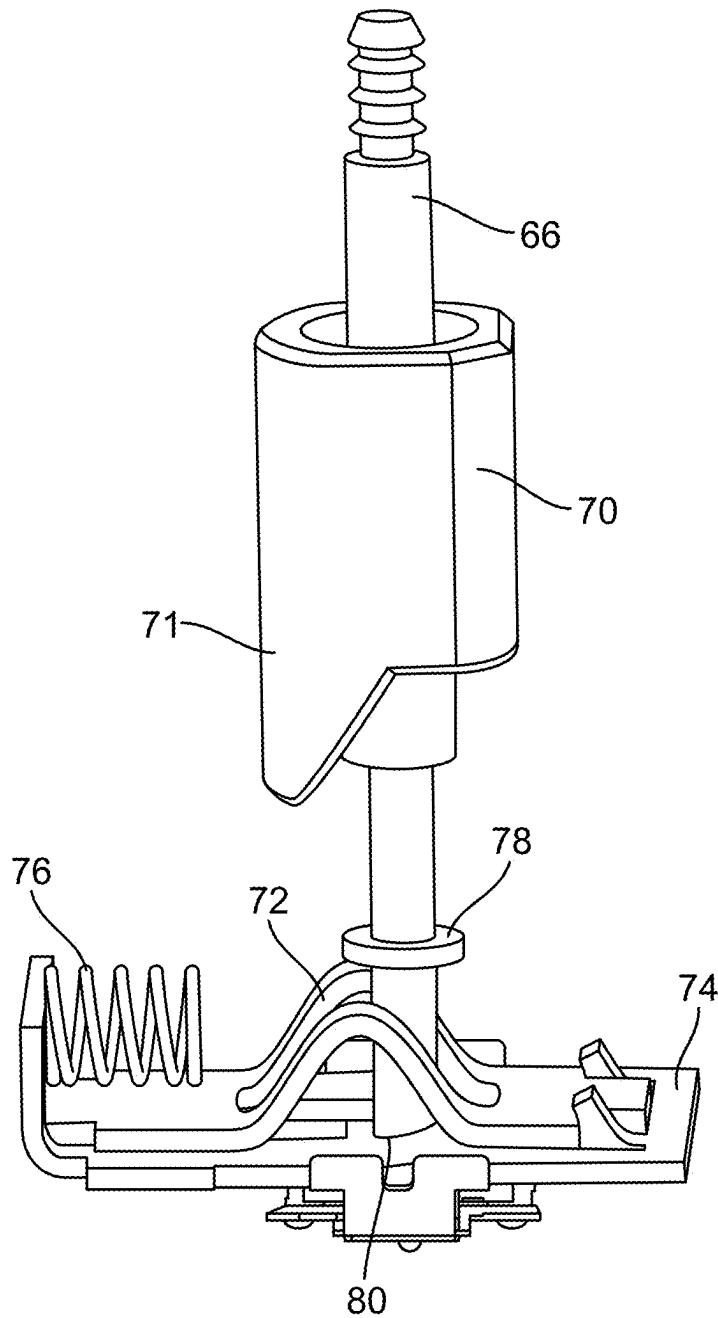


FIG. 14B

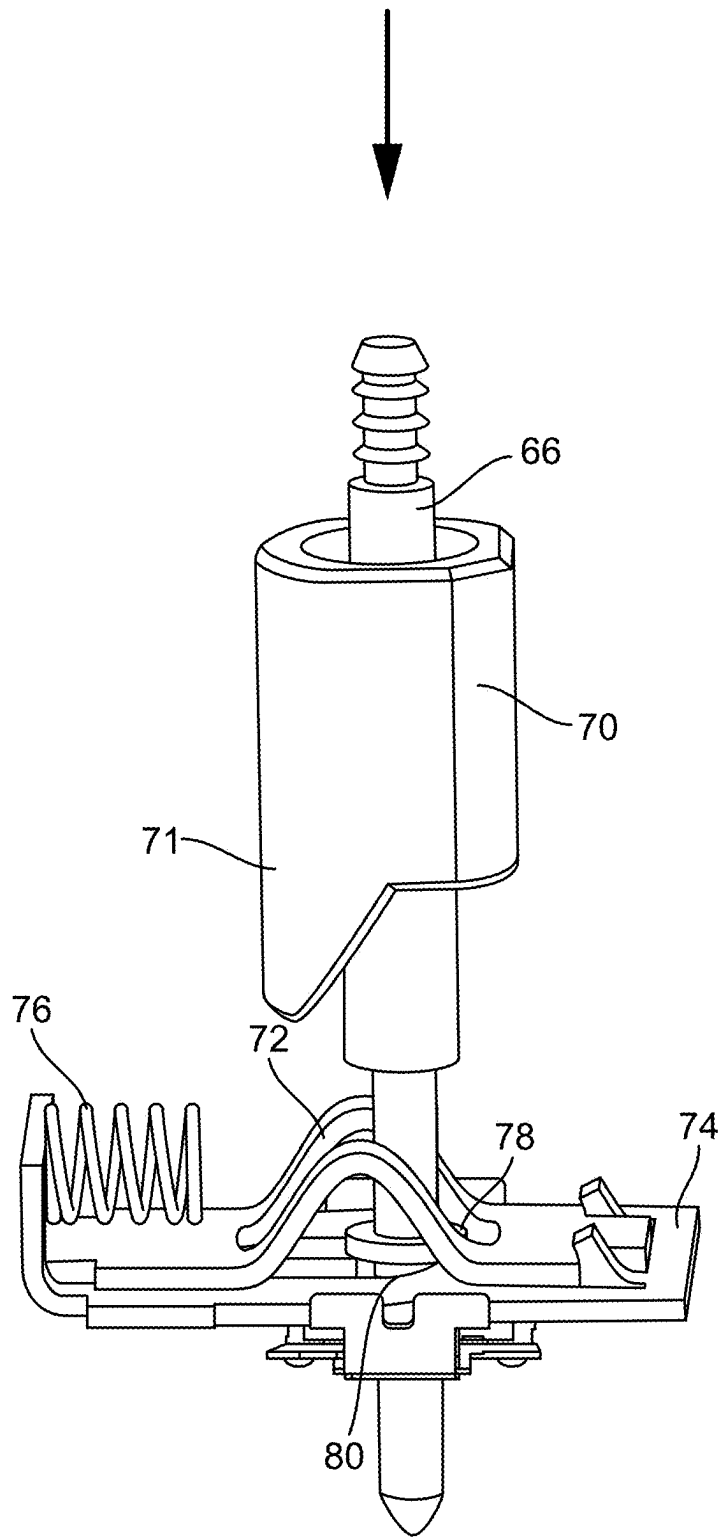


FIG. 15

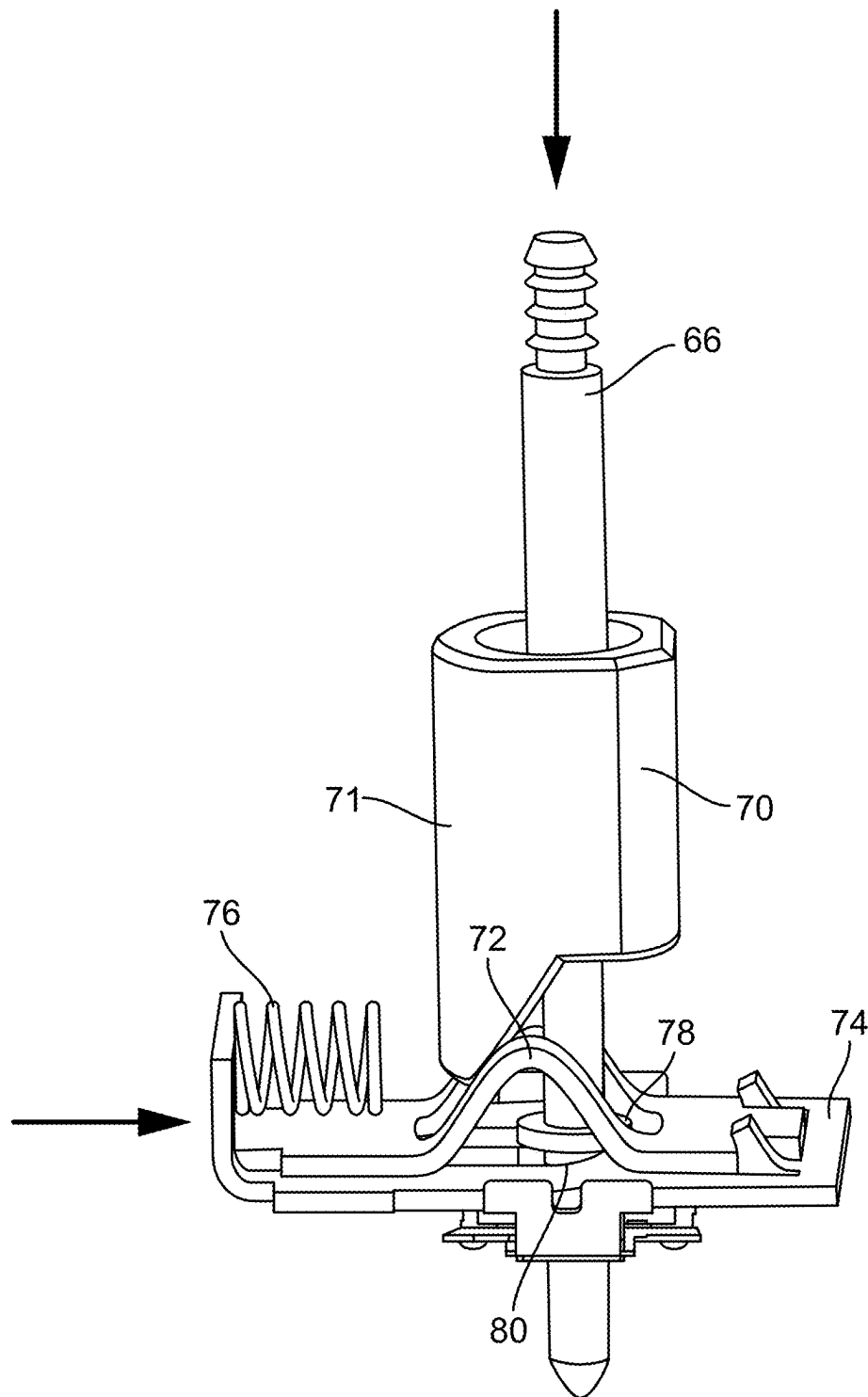


FIG. 16

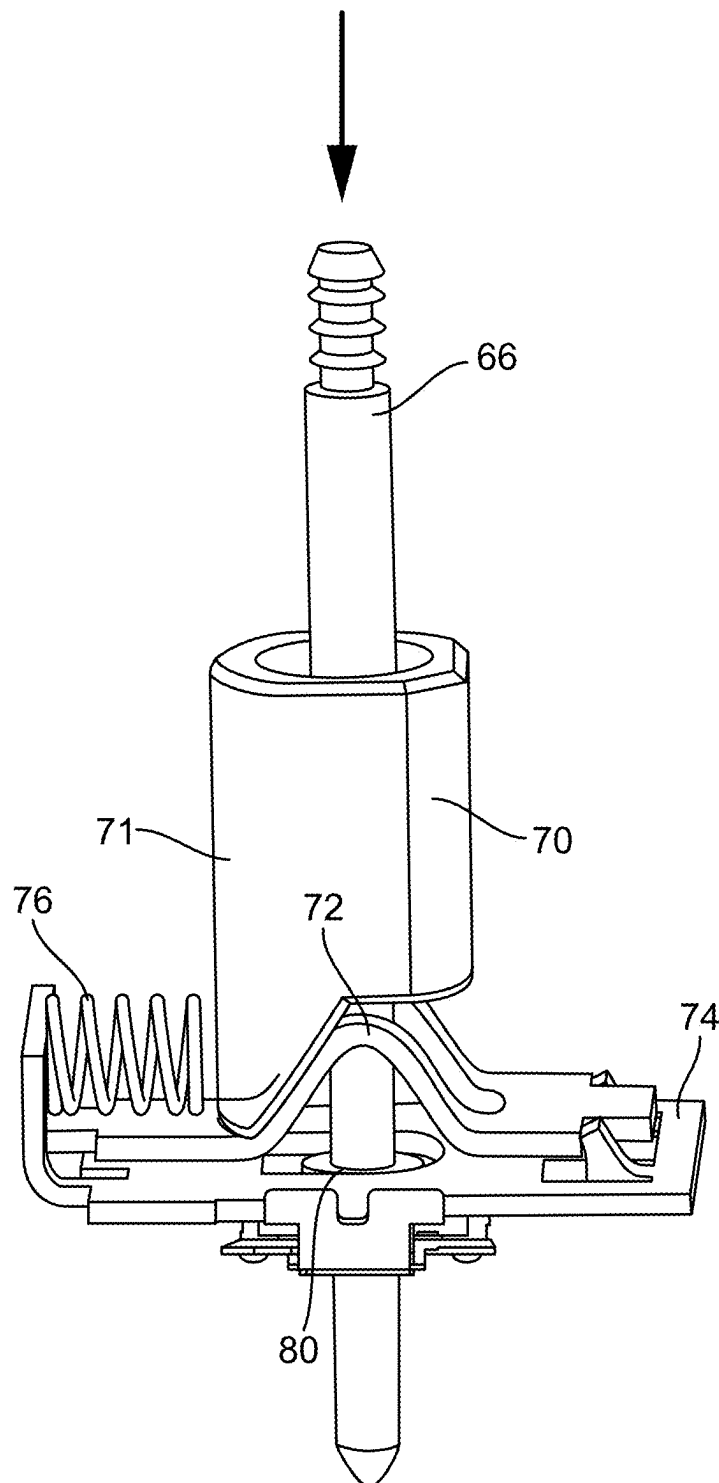


FIG. 17

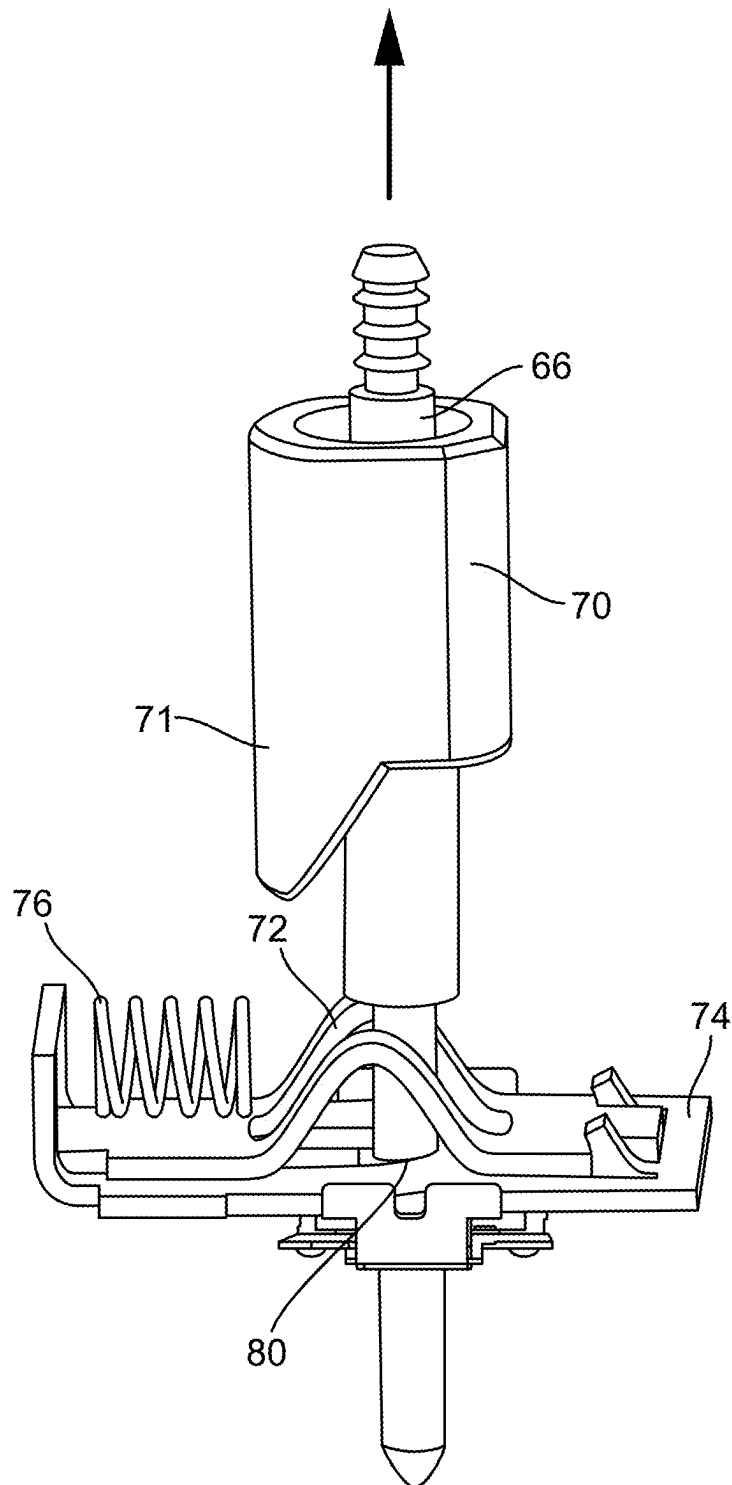


FIG. 18

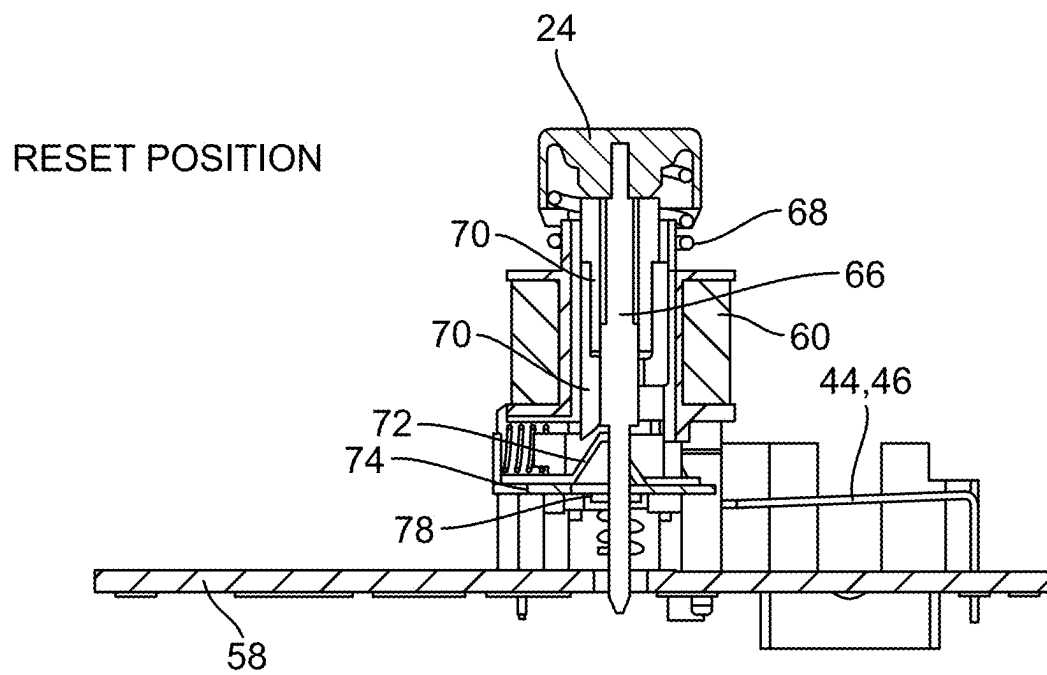


FIG. 19

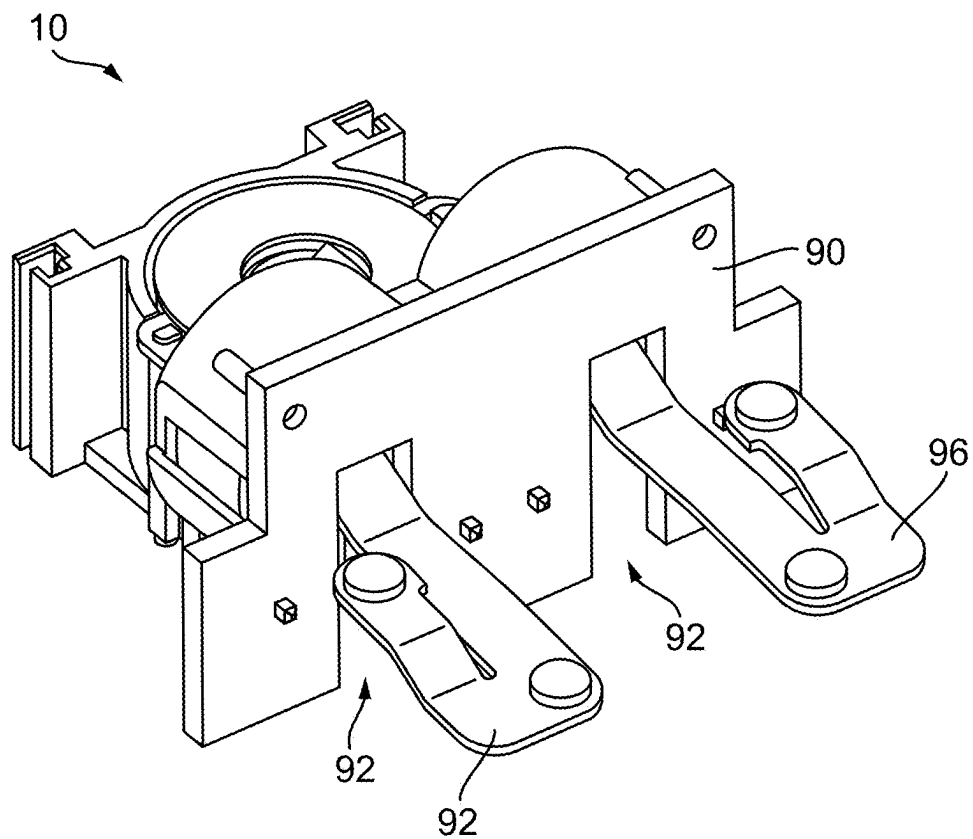


FIG. 20

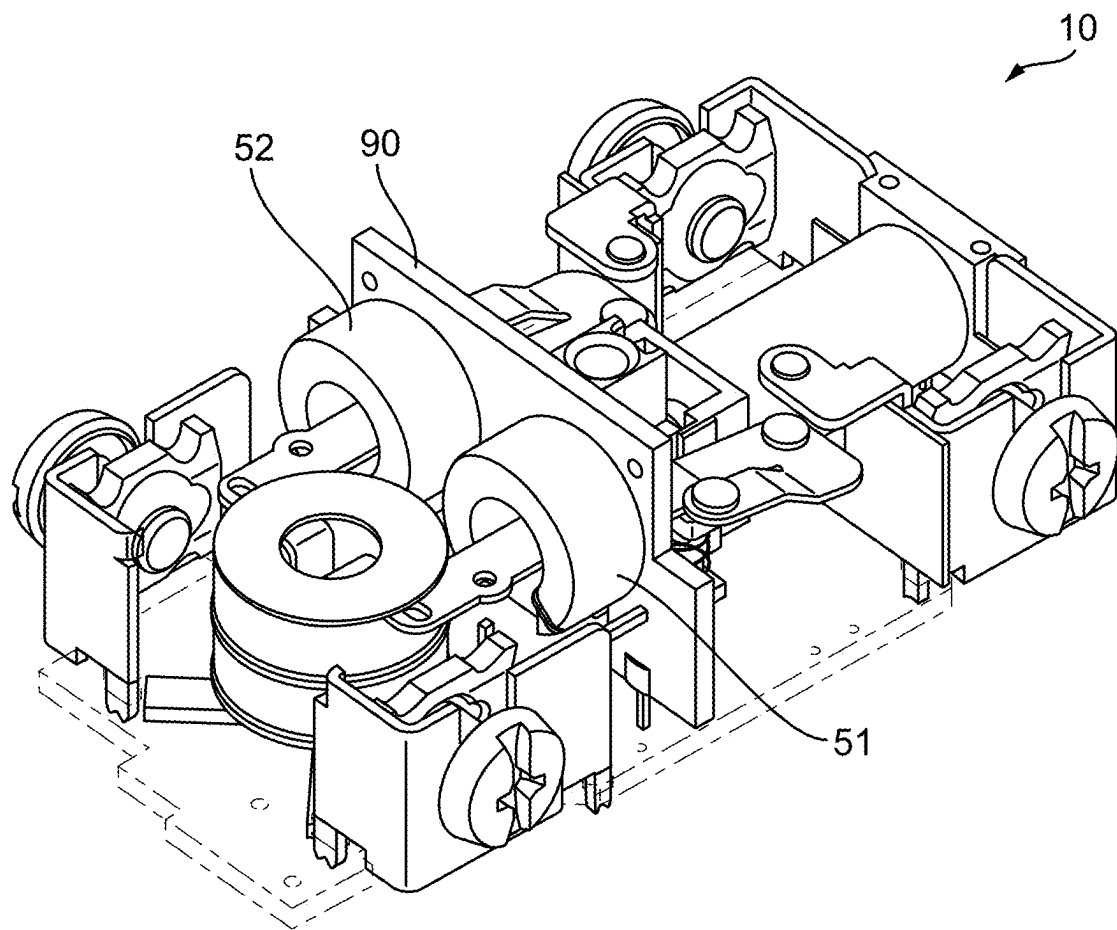


FIG. 21

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SHALLOW ELECTRICAL PROTECTION DEVICE (GFCI, AFCI, AND AFCI/GFCI) SYSTEM AND METHOD

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims priority to U.S. patent application Ser. No. 18/096,813, filed Jan. 13, 2023, which claims priority to U.S. patent application Ser. No. 17/698,690, filed Mar. 18, 2022, which claims priority to, and is a divisional of, U.S. patent application Ser. No. 16/212,141, filed Dec. 6, 2018, which claims priority to U.S. Provisional Patent Application No. 62/595,760, filed Dec. 7, 2017, the entire contents of both of which are expressly incorporated herein by reference.

This application contains subject matter related to subject matter contained in U.S. Pat. No. 8,830,015 B2 entitled, "COMPACT LATCHING MECHANISM FOR SWITCHED ELECTRICAL DEVICE," by Kenny Padro et al., which is assigned to the assignee hereof, and the entire contents of which are expressly incorporated herein by reference.

FIELD

Embodiments relate to switched electrical devices, more particularly to circuit interrupting devices.

SUMMARY

Circuit interrupting devices, such as ground fault circuit interrupter (GFCI) devices, switch to a "tripped" or unlatched state from a "reset" or latched state when one or more conditions is detected. GFCI devices having contacts that are biased toward the open position require a latching mechanism for setting and holding the contacts in a closed position. Likewise, switched electrical devices having contacts that are biased toward the closed position require a latching mechanism for setting and holding the contacts in an open position. Examples of conventional types of devices include devices of the circuit interrupting type, such as circuit breakers, arc fault interrupters, and GFCIs, to name a few.

Many electrical receptacles have built-in ground fault protection circuitry, i.e., GFCI receptacles. Such protection circuitry and the associated mechanisms normally take up a substantial amount of the physical space within a receptacle housing, the size of which is limited by the standard junction boxes in which they must fit. The embodiments disclosed in the present application attempt to solve these problems by providing more compact devices, allowing for shallower receptacles and more space for other elements and/or features.

One embodiment discloses an electrical outlet receptacle comprises a housing including a face plate and a plurality of sensing cores each configured to receive a current flow through a center cavity. The current flow defines a current flow direction through the center cavity, wherein the current flow direction is parallel to the face plate, and the plurality of sensing cores are placed symmetrically in a translational direction across the electrical outlet receptacle.

Another embodiment discloses an electrical outlet receptacle comprises a housing including a face plate and a sensing core configured to receive a current flow through a center cavity. The current flow defines a current flow direc-

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tion through the center cavity of the sensing core, wherein the current flow direction is parallel to the face plate.

Another embodiment discloses an electrical outlet receptacle comprises a circuit board defining a first plane, a set of fixed contacts, a set of movable contacts, a solenoid having a central axis perpendicular to the first plane, a carriage movable axially along the solenoid and configured to interact with the set of movable contacts, a lifting shelf slidably coupled to a slot in the carriage and movable in a translational direction perpendicular to the central axis of the solenoid, a slide mechanism coupled to the lifting shelf and movable in the translational direction of the lifting shelf, a reset plunger with a portion extending through a first end of the solenoid and axially movable therein, and an armature movable axially along the portion of the reset plunger extending through the solenoid. The circuit board includes at least one contact pad. The solenoid includes a second end opposite the first end. The carriage is adapted to advance the set of movable contacts to form electrical communication with the set of fixed contacts during resetting of the electrical outlet receptacle. The lifting shelf includes a latching portion. The slide mechanism includes a cam surface to transform a downward force to a translational force applied to the coupled lifting shelf. The reset plunger includes an intermediate collar configured to engage to the latching portion of the lifting shelf. The armature includes a slanted projection configured to contact the cam surface of the slide mechanism and provide the downward force on the cam surface.

Other aspects of the application will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The aspects and features of various exemplary embodiments will be more apparent from the description of those exemplary embodiments taken with reference to the accompanying drawings, in which:

FIG. 1 is a front perspective view of a receptacle incorporating the resettable switching apparatus of the application;

FIG. 2 is a front perspective view of the receptacle of FIG. 1, with the front and rear covers and tamper-resistant mechanisms removed;

FIG. 3 is a front perspective view showing a configuration of a set of movable contacts and a set of fixed contacts according to one embodiment;

FIG. 4A is a front perspective view of the carrier assembly of the receptacle according to one embodiment;

FIG. 4B is a front perspective view of the carrier assembly of the receptacle according to another embodiment;

FIG. 5A is a front perspective view of the core assembly of the receptacle;

FIG. 5B is a front perspective view showing a configuration of a sense transformer core according to one embodiment;

FIG. 5C is a front perspective view showing a configuration of a sense transformer core according to another embodiment;

FIG. 6A is a side perspective view of the core assembly of the receptacle according to one embodiment;

FIG. 6B is a side perspective view of the core assembly of the receptacle according to another embodiment;

FIG. 7 is an exploded front perspective view of a solenoid assembly according to one embodiment;

FIG. 8 is an assembled front perspective view of the solenoid assembly of FIG. 7;

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FIG. 9 is a side perspective view of the reset plunger assembly and carriage of the solenoid assembly according to one embodiment;

FIG. 10 is another side perspective view from a different angle of the reset plunger assembly and carriage similar to FIG. 9;

FIG. 11 is a side perspective view of the reset plunger assembly and latching mechanism according to one embodiment;

FIG. 12A shows a two-piece latching mechanism design according to one embodiment of the present application;

FIGS. 12B-12C show a one-piece latching mechanism design according to one embodiments of the present application;

FIG. 13 is a bottom perspective view of the solenoid assembly of FIG. 7;

FIG. 14A is a side perspective view of the solenoid assembly in a resting position;

FIG. 14B is a side perspective view of the reset plunger assembly and latching mechanism of the solenoid assembly shown in FIG. 14A;

FIGS. 15-18 are side perspective views of the reset plunger assembly and latching mechanism in progressive states during the resetting process; and

FIG. 19 is a cross-sectional view of the solenoid assembly in a reset position.

FIG. 20 is a perspective view of a printed circuit board and coils of a receptacle according to some embodiments.

FIG. 21 is a perspective view of the printed circuit board and coils of FIG. 20 coupled to a manifold of a receptacle according to some embodiments.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or carried out in various ways.

As described herein, terms such as “front,” “rear,” “side,” “top,” “bottom,” “above,” “below,” “upwardly,” and “downwardly” are intended to facilitate the description of the electrical receptacle of the application, and are not intended to limit the structure of the application to any particular position or orientation.

Exemplary embodiments of devices consistent with the present application include one or more of the novel mechanical and/or electrical features described in detail below. Such features may include a compactly positioned sensing core, a vertical solenoid, and a latching mechanism including a lifting shelf, a slide mechanism, an intermediate collar, and a cam surface. In some exemplary embodiments of the present application, multiple features listed above are incorporated into one element whereas in other exemplary embodiments, each feature is distinct from one another and coupled to interact with each other. The novel mechanical and/or electrical features detailed herein efficiently utilize the space within the device housing to provide more area for additional features and/or components.

FIG. 1 illustrates a perspective view of a GFCI receptacle 10 according to one embodiment of the application. The GFCI receptacle 10 includes a front cover 12 having an outlet face 14 with phase 16, neutral 18, and ground 20 openings. The outlet face 14 also has a central opening 22

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for a reset button 24 adjacent to an opening 26 for a test button 28. Rear cover 36 is secured to front cover 12 by screws (not shown or enumerated). Screw terminals 38 mechanically and/or electrically couple wires when wiring the receptacle 10. A ground yoke/bridge assembly 40 includes standard mounting ears 42 that protrude from the ends of the receptacle 10.

Referring to FIG. 2, the GFCI receptacle 10 with the front cover 12, rear cover 36, and tamper-resistant mechanisms (not enumerated) removed shows phase terminal 30, neutral terminal 32, ground terminal 34, and a circuit board 58. The phase, neutral, and ground terminals 30, 32, 34 are respectively configured to receive electrical plugs 35 of a connecting electrical device, such as a power cord. The circuit board 58 provides control and physical support for most of the working components of the receptacle 10. The phase and neutral terminals 30, 32 may be movable, supported and energized through bus bars 44, 46, respectively. Bus bars 44, 46 act as cantilevered arms that support a set of contacts 48. As shown in the embodiment of FIG. 3, the set of contacts 48 include a set of movable contacts 48A and a set of fixed contacts 48B. Bus bars 44, 46 respectively serve as cantilevered support for the set of movable contacts 48A while the set of fixed contacts 48B is supported by a carrier assembly 8. This configuration may be reversed or changed in other embodiments of the application not described in detail herein. In various embodiments, an indicator light L may be included in the GFCI receptacle 10 and configured to indicate the state of the GFCI receptacle 10.

The resiliency of the cantilevered support provided by the bus bars 44, 46 bias the set of movable contacts 48A away from the set of fixed contacts 48B. A latching mechanism including a movable carriage, described in further detail in the following figures, is used to engage with the set of movable contacts 48A, thereby pushing the set of movable contacts 48A in an upward direction to engage the set of fixed contacts 48B in a closed position during resetting of the GFCI receptacle 10. This upward movement of the set of movable contacts 48A also causes corresponding upward movement in the attached phase and neutral terminals 30, 32 closer to the front cover 12 of the receptacle 10. Electricity may then be delivered from an external power source to the receptacle openings 16, 18, 20. In other embodiments, the resiliency of the cantilevered bus bars 44, 46 may bias the set of movable contacts 48A toward the set of fixed contacts 48B, and a latching mechanism may be employed in reverse to engage and hold the set of movable contacts 48A away from the set of fixed contacts 48B in an open position during tripping of the GFCI receptacle 10. The phase and neutral terminals 30, 32 will likewise increase in distance from the front cover 12, thereby prohibiting the flow of electricity between the external power source and the receptacle openings 16, 18, 20. Various embodiments of the latching mechanism may be used by various application designs, the details of each are not disclosed in detail herein.

Referring to FIG. 4A, in addition to providing structural support for the set of fixed contacts 48B, the carrier assembly 8 also provides structural support for a sense transformer core 50 and conductor windings 52, 54. In another embodiment shown in FIG. 4B, the carrier assembly 8 may provide structural support for multiple sets of sense transformer cores 50, 51, as described in further detail below. Various placements of sense transformer core(s) 50, 51 may be possible and will be further described in the following figures.

FIG. 5A illustrates a perspective view of a core assembly 2 of the GFCI receptacle 10 depicted in FIG. 1. A solenoid

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60 is oriented to define a central axis A. Multiple sense transformer cores 50 may be stacked together and configured to receive a phase conductor winding 52 and a neutral conductor winding 54 through a common central cavity 56. Additional sets of stacked sense transformer cores 51 may be added to the carrier assembly 8 (see FIG. 4B) to provide further measurements, such as arc fault measurements, to the GFCI receptacle 10. The phase and neutral conductor windings 52, 54 respectively direct AC current from the phase and neutral terminals 30, 32 through the central cavity 56, where the current may be measured for potential ground faults or arc faults. The AC current flow through the central cavity 56 defines a direction B, which is perpendicular to the central axis A of the solenoid 60. In the embodiment of FIG. 5A, the two sets of sense transformer cores 50, 51 are placed symmetrically at two ends of the circuit board 58 with current flow directions parallel to each other. This symmetrical placement allows less or essentially no interference of the sense transformer cores 50, 51 with the phase, neutral, or ground openings 16, 18, 20, respectively. It would be appreciated by those skilled in the art that other positioning configurations of the sets of sense transformer cores may be possible and not exhaustively described herein. For example, the current flow directions defined by multiple sets of sense transformer cores 50, 51 may be at an angle to each other and both parallel to the circuit board 58. The angle defined by the current flow directions may be acute, right, or obtuse. In another example shown in FIGS. 5B-C, only one sense transformer core 50 may be included in the GFCI receptacle 10. The sense transformer core 50 may be placed at either ends of the circuit board 58 and with various orientations to allow less or essentially no interference with the phase, neutral, and ground openings 16, 18, 20.

Referring to FIGS. 6A-B, the solenoid 60 is coupled to a carriage 62 that is axially movable along the solenoid 60. On one side, the carriage 62 is coupled to a set of carriage springs 64, the compression force of which distances the carriage 62 from the circuit board 58 in a rest position. On the other side, the carriage 62 is configured to engage the set of movable contacts 48A, which presses down on the carriage 62 when in an unbiased resting position. During the resetting process of the GFCI receptacle 10, the carriage 62 will oppose the resiliency of the abutting set of movable contacts 48A to advance the set movable contacts 48A in an upward direction and form electrical communication with the set of fixed contacts 48B. The upward movement of the set of movable contacts 48A stops once electrical communication is formed with the set of fixed contacts 48B. During the tripping process of the GFCI receptacle 10, the resiliency of the abutting set of movable contacts 48A pushes the carriage 62 in a downward direction back to its original rest position, thereby effectively breaking the electrical connection between the set of movable contacts 48A and the set of fixed contacts 48B. The downward movement range of the set of movable contacts 48A is limited by a stopping plane in the solenoid support structure 61. Once the set of movable contacts 48A hits the stopping plane or returns to the unbiased resting position, push force is no longer exerted on the carriage 62, thereby effectively halting the downward movement and limiting the maximum range of movement of the carriage 62. Resetting and latching of the GFCI receptacle 10 may be controlled by the circuit board 58 that receives ground fault and arc fault signal inputs from the sense transformer cores 50, 51.

FIG. 7 shows an exploded view of a solenoid assembly 4 of the GFCI receptacle 10 according to one embodiment of the present application. The solenoid assembly 4 includes a

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reset button 24, a reset spring 68, a solenoid 60, a reset plunger assembly 6, a solenoid support structure 61, and a circuit board 58. In some embodiments, the solenoid support structure 61 is coupled to the circuit board 58 and supports the solenoid 60. When assembled as shown in FIG. 8, the reset button 24 is biased away from the solenoid 60 via the reset spring 68 as long as no push force is exerted on the reset button 24. When a push force is exerted and subsequently released on the reset button 24, the compression force of the reset spring 68 returns the reset plunger 66 and the reset button 24 to an original resting position biased away from the solenoid support structure 61. Likewise, without an externally exerted downward force, the carriage 62 is biased away from the circuit board 58 via the set of carriage springs 64. The compression force of the carriage springs 64 returns the carriage 62 to an original position biased away from the circuit board 58 when external forces are removed.

Referring to FIGS. 9-11, the reset plunger assembly 6 includes a reset plunger 66 with an intermediate collar 78 and an armature 70 that is axially movable along the length of the reset plunger 66. The armature 70 contains a slanted projection feature 71 that is energized by the solenoid 60 through which the armature 70 extends. The slanted projection feature 71 is configured to engage with the latching mechanism, which is structurally supported by the carriage 62 and the set of carriage springs 64. The latching mechanism includes a cam surface 72 coupled to a lifting plate 74. The lifting plate 74 is coupled through a slot 75 in the carriage 62, as shown in FIG. 10. The lifting plate 74 includes a latching portion 80 configured to receive and engage the intermediate collar 78 of the reset plunger 66 during resetting and tripping of the GFCI receptacle 10, as shown in FIG. 11. A return spring 76 is coupled to one end of the lifting plate 74 and is configured to apply a compression force against one side of the carriage 62.

Two exemplary embodiments of the latching mechanism are shown in FIGS. 12A-C. In the two-piece latching mechanism design of FIG. 12A, the cam surface 72A is configured as a separate triangular plate coupled to channels (not enumerated) in the lifting plate 74A. Additionally, a set of tabs 73A at one end of the lifting plate 74A is configured to engage with edges of the cam surface 72A to transfer a translational force from the cam surface 72A to the lifting plate 74A. The return spring 76A is positioned between the other end of the lifting plate 74A and one side of the carriage 62A. The latching portion 80A is configured as the only opening in the lifting plate 74A and receives/engages with the intermediate collar 78.

In the one-piece latching mechanism of FIGS. 12B-C, the cam surface 72B is integrated into the lifting plate 74B as one element. Since the cam surface 72B does not move independent of the lifting plate 74B, coupling mechanism including channels (not enumerated) and tabs 74A are not necessary in the lifting plate 74B. The lifting plate 74B includes an opening 77B and a latching portion 80B. The return spring 76B is situated in the opening 77B and exerts a compression force between edges of the opening 77B and one side of the carriage 62B. The latching portion 80B is configured to receive/engage with the intermediate collar 78. It would be appreciated by those skilled in the art that other design possibilities not detailed herein may serve to achieve essentially the same results and do not deviate from the teachings of the present application.

According to one embodiment shown in FIG. 13, a contact spring 82 is coupled to the bottom of the lifting plate 74. When in an unlatched pushing state as described in

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further detail below, the contact spring **82** will form electrical communication with at least one contact pads (not shown or enumerated) on the circuit board **58**. This electrical communication will provide a communication signal and power from the circuit board **58** to the solenoid **60**, thereby energizing the armature **70** and resetting the GFCI receptacle **10**.

The GFCI receptacle **10** according to embodiments of the present application has four different states: 1) unlatched state or tripped state, 2) unlatched pushing state, 3) latched pulling state, and 4) latched state or reset state. During the tripped state of FIGS. **14A-B**, the carriage **62** is in a resting position biased away from the circuit board **58** via carriage springs **64**, so the contact spring **82** (not shown) does not form electrical communication with at least one contact pads (not shown or enumerated) on the circuit board **58**. The set of movable contacts **48A** does not engage with the set of fixed contacts **48B** (not shown), and the receptacle terminals **30, 32, 34** remain biased away from the receptacle openings **16, 18, 20** via cantilevered bus lines **44, 46**. Therefore, the solenoid **60** does not receive external power and is not energized, causing the slanted projection feature **71** of the armature **70** to bias away from cam surface **72** (FIG. **14B**). There is no compression force in the return spring **76**, and the engaging portion **80** of the lifting plate **74** is not aligned to receive the intermediate collar **78** biased away from the lifting plate **74**.

Once a downward pushing force is received on the reset plunger **66** from a user pushing down on the reset button **24**, the GFCI receptacle **10** enters the unlatched pushing state of FIG. **15**. In the unlatched pushing state, the downward force pushes the reset plunger **66** towards the lifting plate **74** until the intermediate collar **78** engages with an upper surface of the engaging portion **80**. Because the engaging portion **80** is misaligned with the intermediate collar **78** from the previous tripped state, the intermediate collar **78** engages with but does not latch to the upper surface of engaging portion **80**. Thus, the downward force from the intermediate collar **78** transfers to the engaging portion **80** and the lifting plate **74**, which results in downward movement of the carriage **62** via the slot **75** (FIG. **10**). This downward movement continues until the contact springs **82** (not shown) form electrical communication with at least one contact pads (not shown or enumerated) on the circuit board **58**. Upon contact, electrical power and communication is sent from the circuit board **58** to the solenoid **60**, energizing the solenoid on a positive half cycle of the input AC power and moving the armature **70** axially along the reset plunger **66**. Referring to FIG. **16**, the slanted projection feature **71** of the armature **70** engages with the cam surface **72**, which translates the downward force to a translational force parallel to the circuit board **58**. Translational movement of the cam surface **72** also translationally moves the coupled lifting plate **74** against the compression force of the return spring **76**, thus aligning the engaging portion **80** with the intermediate collar **78**. Referring to FIG. **17**, the continued downward force on the reset plunger **66** applied by the user causes the intermediate collar **78** to travel through the aligned engaging portion **80**. At this point, the solenoid **60** de-energizes on a negative half cycle of the input AC power and retracts axially along the reset plunger **66**, as shown in FIG. **18**. The compression force of the return spring **76** pushes the side of the carriage **62** and returns the lifting plate **74** and cam surface **72** back to the original position. In this original position, the intermediate collar **78** is once again misaligned with the engaging portion **80**. When the user releases the downward pushing force on the reset plunger **66**, the reset spring **68** provides an upward

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pulling force on the reset plunger **66** and intermediate collar **78**, thereby latching and locking the intermediate collar **78** to a lower surface of the engaging portion **80**. Hence, the GFCI receptacle **10** enters the latched pulling state of the resetting process.

When the GFCI receptacle **10** is in the latched pulling state shown in FIG. **19**, the compression force of the reset spring **68** creates an upward force on the reset button **24** and the coupled reset plunger **66**. This upward force pulls the intermediate collar **78** along with the latched lifting plate **74**, which is coupled to the carriage **62** via the slot **75**, causing the carriage **62** to move axially upward along the solenoid **60**. The axially upward movement of the carriage **62** opposes the resiliency of the abutting set of movable contacts **48A** and disconnects the contact springs **82** (not shown) from the at least one contact pads (not shown or enumerated) on the circuit board **58**, thus preventing continued energizations of the solenoid **60**. The carriage **62** engages with the set of movable contacts **48A** to form electrical connection with the set of fixed contacts **48B**. Correspondingly, the receptacle terminals **30, 32, 34** also resist the cantilevered bus lines **44, 46** and move closer to the front cover **12**. Once electrical communication between the set of movable contacts **48A** and the set of fixed contacts **48B** is formed, electricity may be delivered from the receptacle terminals **30, 32, 34** to the receptacle openings **16, 18, 20** via the bus lines **44, 46**. Hence, the GFCI receptacle **10** is fully reset.

When the sense transformer cores **50, 51** detect the present of a fault, the GFCI receptacle **10** completes a tripping process. During the tripping process, the GFCI receptacle **10** experiences the states of the resetting process in reverse order, thereby unlatching the intermediate collar **78** from the latching portion **80** and breaking the electrical communication between the set of movable contacts **48A** and the set of fixed contacts **48B**.

FIGS. **20 & 21** illustrate a GFCI receptacle **10** according to some embodiments. In the illustrated embodiment, the GFCI receptacle **10** includes a printed circuit board **90**. In some embodiments, the printed circuit board **90** includes one or more slots, or apertures, **92**. As illustrated, the slots **92** may be configured receive, or be placed over, line conductors **94** and/or the neutral conductors **96**, or a portion thereof (for example, bus bars **44, 46**). The printed circuit board **90** may further include, or be coupled to, coils (for example, transformer cores **50, 51**), which may be used to sense and/or monitor a current. In such an embodiment, the coils may also include a slot, aperture, configured to receive, or be placed over, the line conductors and/or the neutral conductors, or a portion thereof (for example, bus bars **44, 46**).

In certain other embodiments, additional elements, such as springs, contacts, etc., may be included in various places within the GFCI receptacle **10** to accomplish resetting or tripping of the device. All combinations of embodiments and variations of design are not exhaustively described in detail herein. Said combinations and variations are understood by those skilled in the art as not deviating from the teachings of the present application.

What is claimed is:

1. An electrical outlet receptacle comprising:
 - a housing including a face plate; and
 - a solenoid having a central axis perpendicular to the face plate;
 - a plurality of sensing cores each configured to receive a current flow through a center cavity, the current flow defining a current flow direction through the center cavity,

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wherein the current flow direction is parallel to the face plate; and

wherein the plurality of sensing cores are placed symmetrically in a translational direction across the electrical outlet receptacle.

2. The electrical outlet receptacle of claim 1, further comprising:

a set of fixed contacts and a set of movable contacts;

a solenoid having a central axis perpendicular to the face plate, the solenoid having a first end and a second end opposite the first end;

a carriage movable axially along the solenoid and configured to interact with the set of movable contacts, the carriage configured to advance the set of movable contacts to form electrical communication with the set of fixed contacts during resetting of the electrical outlet receptacle;

a reset plunger with a portion extending through the first end of the solenoid and axially movable therein, the reset plunger being biased toward a rest position away from the carriage; and

an armature movable axially along the portion of the reset plunger extending through the solenoid, the armature being biased toward a rest position away from the carriage,

wherein the reset plunger includes an engaging portion and the carriage includes a latching portion, the engaging portion of the reset plunger being configured to connect to the latching portion of the carriage.

3. The electrical outlet receptacle of claim 2, wherein the solenoid includes a coil bobbin with an inner coil and an outer coil.

4. The electrical outlet receptacle of claim 2, wherein the reset plunger further includes a spring configured to bias the reset plunger toward the rest position.

5. The electrical outlet receptacle of claim 2, wherein the armature interacts with the latching portion of the carriage when the solenoid is momentarily energized to cause engagement of the engaging portion to the latching portion in a latched state.

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6. The electrical outlet receptacle of claim 5, wherein in the latched state, the armature interacts with the latching portion of the carriage when the solenoid is momentarily energized to cause disengagement of the engaging portion from the latching portion in an unlatched state.

7. The electrical outlet receptacle of claim 5, wherein pressing of the reset plunger causes the engaging portion to apply a downward force on the carriage at least until the electrical outlet receptacle is in the latched state, and subsequent release of the reset plunger causes the engaging portion to apply an upward force on the carriage to close the set of movable contacts and the set of fixed contacts.

8. The electrical outlet receptacle of claim 2, further comprising at least one sensing core configured to receive a current flow through a center cavity, the current flow defining a current flow direction through the center cavity.

9. The electrical outlet receptacle of claim 8, wherein the at least one sensing core includes a ground fault circuit interrupter (GFCI) core and an arc fault circuit interrupter (AFCI) core, the GFCI core having a first current flow direction and the AFCI core having a second current flow direction.

10. The electrical outlet receptacle of claim 9, wherein the first current flow direction and the second current flow direction are identical to each other.

11. The electrical outlet receptacle of claim 9, wherein the first current flow direction and the second current flow direction are substantially parallel to each other.

12. The electrical outlet receptacle of claim 9, wherein the first current flow direction and the second current flow direction are at an oblique angle to each other.

13. The electrical outlet receptacle of claim 9, wherein the first current flow direction and the second current flow direction are at a right angle to each other.

14. The electrical outlet receptacle of claim 9, wherein the GFCI core and AFCI core are positioned in a symmetrical placement across a translational direction of the electrical outlet receptacle, the symmetrical placement allowing for no interference of the GFCI core and the AFCI core with contact terminals connectable to an external device.

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